

# Agent Automation Creator (AAC) v1.1

## Process-First Framework for AI-Augmented Workflow Design

**Status:** v1.1 **Owner:** Ankit Patel **Internal name:** Agent Automation Creator (AAC). Final commercial name TBD. **Supersedes:** AAC v1.0 (formerly "MSP-FLOW v1.0"), all prior v0.x drafts **Pressure-tested against:** After-hours intake (Replace mode, high volume, low stakes per run) and Rx fax → RevolutionEHR entry (Augment mode, low volume, high stakes per run)

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## 0. Frame and Philosophy

### 0.1 The core claim

A workflow is a substrate. AI is one of four possible runtimes that sit on top of it. Most failed AI projects design the AI first and try to fit a process around it. AAC reverses the order: map the process correctly, then assign the right runtime to each piece of work, then apply closed-loop AI discipline only where AI is the assigned runtime.

### 0.2 The three layers

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LAYER 3: Closed-Loop AI Framework (applied only to AI-assigned work)
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LAYER 2: Runtime Assignment (per work element: D / C / A / H)
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LAYER 1: Process Map (the substrate, three levels of granularity)
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You build bottom-up. You can't skip a layer.

### 0.3 Non-negotiable principles

1. **Process before automation.** Layer 1 is drawn without reference to AI capabilities.
2. **Atomic decisions over composite ones.** Every decision is its own work element.
3. **AI for judgment, code for action.** AI produces structured output. Deterministic code executes the action.

4. **Refusal is a first-class outcome.** Every AI work element has an explicit refusal path.
5. **The spec is the artifact.** A workflow exists when its spec is complete.
6. **Constraints come from the user's domain, not the framework.** AAC defines constraint shapes. The user's domain populates them.

## 0.4 What's new in v1.1

| # | Change                                                              | Type       |
|---|---------------------------------------------------------------------|------------|
| 1 | Confidence thresholds default by action consequence                 | Refinement |
| 2 | Cross-check conditionality default by action consequence            | Refinement |
| 3 | Graduation criteria scale with per-error cost band (domain-neutral) | Refinement |
| 4 | Hard-refuse policy classes as new BOUNDED sub-section               | Addition   |
| 5 | Regulatory exposure splits into 4 constraint dimensions             | Addition   |
| 6 | Terminal H vs migration-eligible H with reason class                | Addition   |
| 7 | Value mode (Replace / Augment / Extend) required field              | Addition   |
| 8 | Rubric expands from 51 to 57 items                                  | Addition   |

All changes are domain-neutral. They work in healthcare, finance, legal, manufacturing, and other domains. The framework defines what types of constraints exist; the user's domain populates the specific values.

## Part I — Process Mapping Discipline

### 1.1 The three-level hierarchy

| Level               | Definition                                                             | Analog (Lean)           | Analog (Software) |
|---------------------|------------------------------------------------------------------------|-------------------------|-------------------|
| <b>Work Element</b> | Atomic, indivisible unit. One verb, one input class, one output class. | Therblig / work element | Pure function     |
| <b>Node</b>         | Logical group of work elements sharing context,                        | Workstation             | Service or        |

|                |                                                                                      |                 |                    |
|----------------|--------------------------------------------------------------------------------------|-----------------|--------------------|
|                | ownership, transactional boundary                                                    | / operation     | module             |
| <b>Process</b> | Full graph of nodes connected by edges, with triggers, decisions, queues, boundaries | Production line | Distributed system |

Design top-down (process shape). Verify bottom-up (every element has a defined runtime, every node composes correctly, the process produces the promised value).

## 1.2 The 12 canonical work element types

| #  | Type      | What It Does                                       | Default Runtime           |
|----|-----------|----------------------------------------------------|---------------------------|
| 1  | Capture   | Receive input from external source                 | D                         |
| 2  | Validate  | Check input or output against a rule set           | D (rules) or C (judgment) |
| 3  | Transform | Convert form A to form B deterministically         | D                         |
| 4  | Classify  | Assign input to one of N enumerated categories     | C                         |
| 5  | Decide    | Choose between N branches based on input state     | D (rules) or C (judgment) |
| 6  | Generate  | Produce new content from input and context         | C                         |
| 7  | Lookup    | Retrieve from a data source                        | D                         |
| 8  | Score     | Assign a numeric value to an input                 | C or D                    |
| 9  | Compose   | Combine multiple inputs into one output            | D or C                    |
| 10 | Execute   | Perform an external action (API call, write, send) | D                         |
| 11 | Log       | Record state to telemetry                          | D                         |
| 12 | Refuse    | Explicit no-action exit with logged reason         | D                         |

Every work element must be exactly one type. If it isn't, decompose further.

## 1.3 Required fields per work element

Type, input contract (schema), output contract (schema), runtime (D/C/A/H), latency budget (ms), cost budget (per run), failure mode + handler, confidence requirement (C/A only).

## 1.4 Node composition rules

1. Shared context across elements within a node
2. Single transactional boundary (succeeds or fails end-to-end)
3. One named human owner
4. Bounded runtime mix (D and C freely; A or H requires "human-gated" marking)
5. Cross-node communication only via persisted state, never shared working memory

## 1.5 Edge required metadata

Latency budget, volume/day, baseline failure rate, data contract, trigger type (sync/async/scheduled/event).

## 1.6 Anti-patterns

| Anti-pattern                                  | Why it fails                                  |
|-----------------------------------------------|-----------------------------------------------|
| Mega-node (10+ things in one node)            | Impossible to own, gate, or roll back cleanly |
| Implicit decisions ("if it seems urgent...")  | Untestable, ungatable, drifts silently        |
| Hidden queue (DLQ no one inspects)            | Work backs up invisibly                       |
| AI deciding AI scope                          | Violates BOUNDED                              |
| Soft handoffs (implicit context across edges) | Receiving node must guess; failure rate rises |
| No refuse path                                | Forces improvisation under uncertainty        |

## 1.7 Notation

Simplified subset of BPMN 2.0. Anyone trained in BPMN can read it. Avoid BPMN's exotic constructs (compensation events, complex gateways, event subprocesses).

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## Part II — Runtime Assignment

### 2.1 The four runtimes

| Code | Name | Description |
|------|------|-------------|
|      |      |             |

|          |                    |                                                                        |
|----------|--------------------|------------------------------------------------------------------------|
| <b>D</b> | Deterministic Code | Pure function or rule engine. Same input always produces same output.  |
| <b>C</b> | Closed-loop AI     | AI does judgment work; no human in the per-run loop. Bounded by gates. |
| <b>A</b> | Assisted AI        | AI proposes, human reviews/approves before action.                     |
| <b>H</b> | Human Only         | Pure human work. AI may inform but does not decide or act.             |

## 2.2 Attributes that determine runtime

Work elements and nodes carry these attributes. Runtime is read off them, not chosen by preference.

| Attribute                                           | Values                                                                                |
|-----------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Decision Class</b>                               | Routine-rule / Probabilistic / Novel / Value-laden                                    |
| <b>Action Consequence</b>                           | Reversible-cheap / Reversible-costly / Irreversible-bounded / Irreversible-unbounded  |
| <b>Input Type</b>                                   | Structured / Semi-structured / Unstructured / Ambiguous                               |
| <b>Volume × Stakes</b>                              | High-vol low-stakes / High-vol high-stakes / Low-vol low-stakes / Low-vol high-stakes |
| <b>Data Classification</b><br><i>(new in v1.1)</i>  | What sensitivity/category does the data carry? Domain-populated.                      |
| <b>Provider Contract</b> <i>(new in v1.1)</i>       | What contractual constraints govern third-party processors? Domain-populated.         |
| <b>Data Residency</b> <i>(new in v1.1)</i>          | Where can data physically reside or be processed? Domain-populated.                   |
| <b>Retention + Deletion</b><br><i>(new in v1.1)</i> | What lifecycle rules govern this data? Domain-populated.                              |

## 2.3 The four data-handling constraint dimensions (v1.1)

These four together pre-filter the eligible tool stack **before** any other design decision. A tool that fails any of them is excluded at design time, not deployment time.

|  |         |            |           |  |               |
|--|---------|------------|-----------|--|---------------|
|  | What it | Healthcare | Financial |  | Manufacturing |
|--|---------|------------|-----------|--|---------------|

| Dimension            | asks                        | ex           | ex                               | Legal ex                     | ex                               |
|----------------------|-----------------------------|--------------|----------------------------------|------------------------------|----------------------------------|
| Data Classification  | Sensitivity / category      | PHI          | NPI / PCI / MNPI                 | Privileged / work-product    | Trade secret, ITAR               |
| Provider Contract    | Third-party constraints     | BAA required | DPA + sub-processor limits       | Confidentiality agreement    | NDA + export-control clauses     |
| Data Residency       | Where data can live/process | US-only      | EU-only (GDPR), country-specific | Per-engagement               | Country-of-origin                |
| Retention + Deletion | Lifecycle rules             | 7-year HIPAA | 7-year SOX                       | Per matter / litigation hold | Indefinite for product liability |

## 2.4 Assignment rules

### Default-cheapest rule

Assign the cheapest runtime that satisfies all attribute constraints. Cost order:  $D < C < A < H$ .

### Runtime eligibility matrix

| Attribute pattern                     | D | C                | A           | H |
|---------------------------------------|---|------------------|-------------|---|
| Routine-rule + Structured input       | ✓ | ✓                | ✓           | ✓ |
| Probabilistic + Structured/Semi input | × | ✓                | ✓           | ✓ |
| Probabilistic + Unstructured input    | × | ✓ (strong gates) | ✓           | ✓ |
| Novel + any                           | × | ×                | ✓           | ✓ |
| Value-laden + any                     | × | ×                | ×           | ✓ |
| Irreversible-unbounded action         | × | ×                | conditional | ✓ |
| Ambiguous input                       | × | ×                | ✓           | ✓ |

Read horizontally: pick the leftmost ✓ that satisfies the row.

## Constraint dimension override

Even if the eligibility matrix permits C, the runtime is excluded if any tool capable of running it fails one of the four constraint dimensions. Example: if your only available cross-check model isn't BAA-covered and the data is PHI, you can't run C on that element. Downgrade to A until a covered tool exists.

## 2.5 Migration paths and terminal H (v1.1)

Every H element must declare its **terminal status** and, if terminal, its **reason class**.

| H type                      | Definition                                            | Migration target                            |
|-----------------------------|-------------------------------------------------------|---------------------------------------------|
| <b>Migration-eligible H</b> | Human in loop due to current eval coverage limits     | A → C as evidence accumulates               |
| <b>Terminal H</b>           | Human in loop by policy, regulation, or risk decision | Never migrates by default; quarterly review |

### Terminal H reason classes (5)

| Reason                  | Definition                                               | Cross-domain examples                                     |
|-------------------------|----------------------------------------------------------|-----------------------------------------------------------|
| <b>Regulatory</b>       | Law requires human decision                              | DEA, fiduciary, judicial signoff, RN/PharmD authorization |
| <b>Contractual</b>      | Customer/partner contract requires human                 | SLA-mandated review, white-glove tier                     |
| <b>Risk Policy</b>      | Internal policy holds human in loop until further notice | High-dollar approval, novel customer type                 |
| <b>Capability Gap</b>   | AI not yet capable enough                                | Standard migration candidate (A → C)                      |
| <b>Cultural / Brand</b> | Customer expects to talk to a person                     | High-touch sales, complaint resolution                    |

Only "Capability Gap" is migration-eligible. The other four are reviewed on quarterly cadence for whether the underlying reason still holds.

## Migration thresholds

| From → To                 | Required evidence                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| H<br>(Capability Gap) → A | Eval set ≥500 cases covering ≥80% of historical distribution; AI/human agreement meets cost-band threshold (see Part V §5.4) |
| A → C                     | 8+ weeks at A; agreement meets cost-band threshold; zero high-severity escapes; failure modes catalogued                     |
| C → D                     | Rare. Only when AI behavior turns out to be expressible as deterministic rules.                                              |
| Any → H<br>downgrade      | Drift, regulatory change, quality regression, contractual change                                                             |

## Part III — The Closed-Loop AI Framework

Applies only to work elements assigned runtime C. For D, no AI discipline needed. For A, a relaxed version applies (human is one of the gate checks). For H, this part doesn't apply.

### 3.1 The five properties

Every C work element must satisfy all five. If any is missing, it isn't closed-loop and must run as A until fixed.

#### 3.1.1 BOUNDED

##### Action vocabulary

- Finite, enumerated, code-defined
- AI selects parameters within a class; AI never selects the action class
- OOD inputs detected → route to refuse
- Test: can you list every action this element will ever take?

##### Hard-refuse policy classes (v1.1)

Every BOUNDED spec must enumerate hard-refuse policy classes: input classes that are excluded from the AI path entirely by policy, regardless of confidence or capability. These are policy-defined, not capability-defined. They bypass the AI path before any gate runs.

Examples populated by domain:



- Healthcare: controlled substances, pediatric prescriptions, behavioral health records
- Financial: wire transfers above threshold, sanctioned-party transactions, ERISA-covered actions
- Legal: attorney-client privileged content, opposing-counsel communications
- Manufacturing: safety-critical control changes, recall-affected SKUs
- Internal policy: executive comp, M&A activity, board materials
- Cross-domain: minors, deceased individuals, named litigants

Hard-refuse events are audit-logged separately from graceful refuse and reviewed quarterly in a compliance audit.

### 3.1.2 GROUNDED

- AI's factual claims anchored to retrieved sources
- Structured output requires source IDs for factual fields
- Validators reject outputs that fail grounding requirements
- Test: can every output field be traced to a source artifact or prompt instruction?

### 3.1.3 GATED (4 gates, all must pass)

| Gate             | Check                                                                           |
|------------------|---------------------------------------------------------------------------------|
| Input gate       | Schema, value ranges, content policy, OOD detector, hard-refuse class detection |
| Output gate      | Schema, policy, grounding, confidence threshold, refusal trigger                |
| Cross-check gate | Second AI (different model family) verifies primary on high-stakes              |
| Action gate      | Idempotency, rate limit, blast radius, pre-execution sanity check               |

### 3.1.4 OBSERVED

- Per-run telemetry: input hash, output, model version, latency, cost, confidence, validator outcomes
- Statistical process control charts on every metric

- Input drift, output drift, confidence drift detected independently
- Champion-challenger on every model or prompt change

### 3.1.5 GOVERNED

- Refuse always available, always cheap, always logged
- Confidence thresholds explicit and version-controlled
- Hard-refuse policy classes enforced before any AI processing
- Ambiguous inputs route to deterministic fallback, never AI improvisation
- Kill switch + circuit breakers per action class

### 3.2 Confidence thresholds scale with action consequence (v1.1)

Minimum confidence required for an AI output to pass the output gate defaults by action consequence. These are floors, not ceilings.

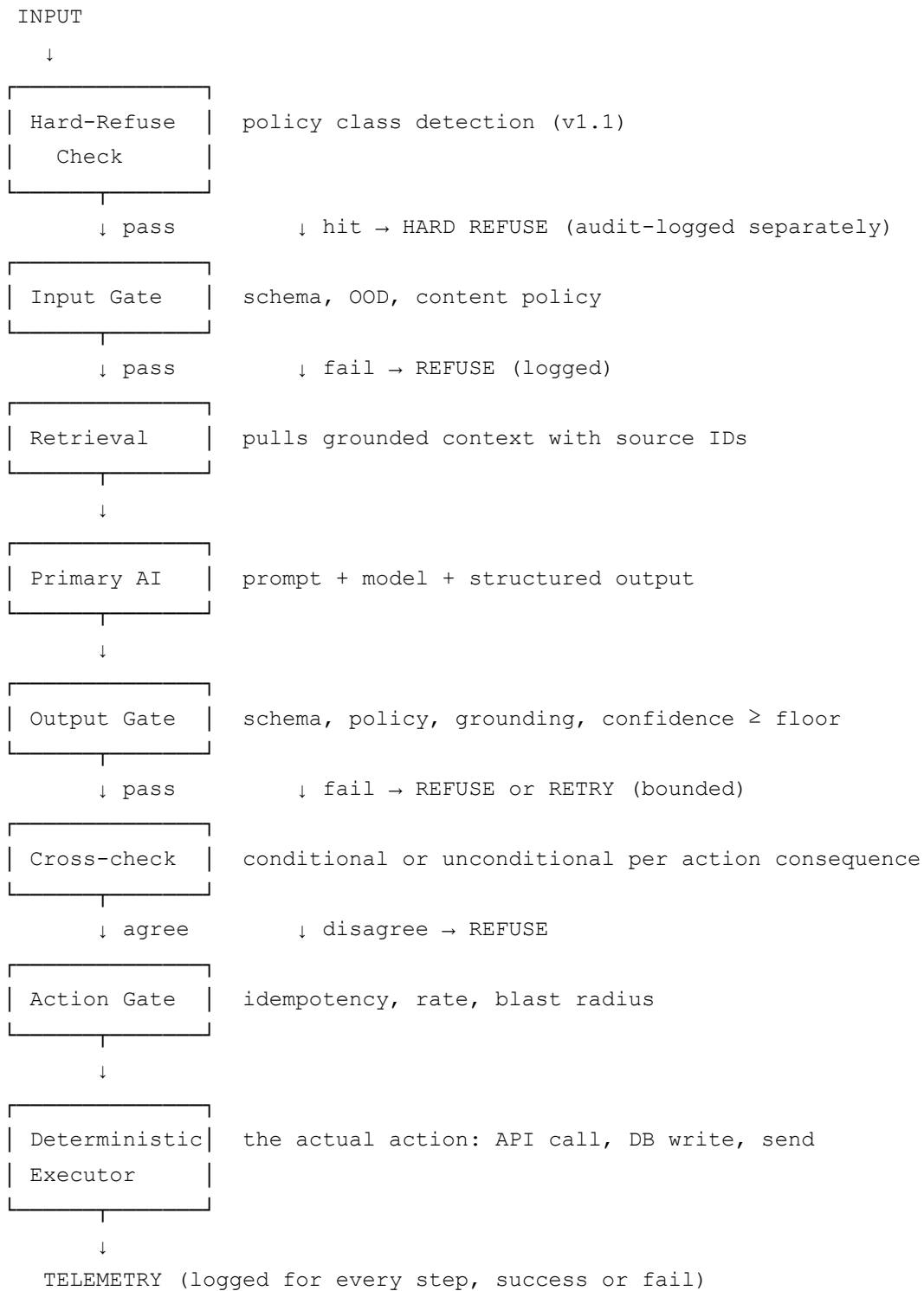
| Action consequence     | Default minimum confidence |
|------------------------|----------------------------|
| Reversible-cheap       | 0.6                        |
| Reversible-costly      | 0.8                        |
| Irreversible-bounded   | 0.92                       |
| Irreversible-unbounded | C not eligible             |

Override allowed only with documented justification in the spec.

### 3.3 Cross-check conditionality scales with action consequence (v1.1)

| Action consequence     | Cross-check policy                                          |
|------------------------|-------------------------------------------------------------|
| Reversible-cheap       | Conditional on low primary confidence                       |
| Reversible-costly      | Conditional on low primary confidence or high stakes signal |
| Irreversible-bounded   | <b>Unconditional</b>                                        |
| Irreversible-unbounded | N/A (C not eligible)                                        |

### 3.4 Reference architecture for a C work element



### 3.5 Cost discipline

| Lever          | Mechanism                                                          | Typical Savings |
|----------------|--------------------------------------------------------------------|-----------------|
| Tiered routing | Cheap model first; expensive only on low-confidence or high-stakes | 60-85%          |

|                          |                                                                                        |                       |
|--------------------------|----------------------------------------------------------------------------------------|-----------------------|
| Prompt caching           | Static system prompts cached at provider level                                         | 30-60%                |
| Right-size retrieval     | Top-K with hard cap                                                                    | 40-70%                |
| Refuse early             | Input gate + hard-refuse check before AI call                                          | 100% on rejected      |
| Conditional cross-check  | Run only when primary confidence below threshold (where allowed by action consequence) | 50-80% on cross-check |
| Push to D where possible | Anything rule-expressible becomes code                                                 | Variable, often huge  |
| Batch where async        | Discount windows on batch-eligible workflows                                           | 30-50%                |
| Quarterly model re-bake  | Re-benchmark cheaper models against eval set                                           | 40-70% over time      |

### 3.6 Drift defense

| Drift Type       | Signal                                                         | Response                                                           |
|------------------|----------------------------------------------------------------|--------------------------------------------------------------------|
| Input drift      | Statistical distance between rolling distribution and baseline | Auto-pause if severe; sample for eval set extension; page engineer |
| Model drift      | Provider silently updates model behind version label           | Pin version; canary test on documented updates                     |
| Output drift     | Distribution of outputs shifts vs baseline                     | Compare against shadow run with prior version                      |
| Confidence drift | Distribution of confidence scores shifts                       | Re-validate eval set is representative                             |
| Spec drift       | Business rules change but workflow doesn't                     | Mandatory re-eval on spec changes; quarterly business-rule audit   |

Responses never include "a human reviews the next 100 runs." Responses are: pause, route to deterministic fallback, page on-call engineer to fix the system offline.

## Part IV — The Spec Document (16 sections)

A workflow exists when this spec is complete. Every section has required fields.

### Section 1 — Work Object

- Noun definition (one sentence)
- Example inputs (3+)
- In-scope boundary
- Out-of-scope boundary (explicit)

### Section 2 — Value Promise (*updated v1.1*)

**Value mode (required, pick one):**

| Mode           | Meaning                                                  | Primary metric                                   |
|----------------|----------------------------------------------------------|--------------------------------------------------|
| <b>Replace</b> | AI takes over work humans were doing                     | Volume of work no longer requiring human         |
| <b>Augment</b> | AI reduces human effort per unit of work                 | % time reduction per unit, quality held constant |
| <b>Extend</b>  | AI enables work that wasn't economically feasible before | New work volume captured                         |

**Four value dimensions:**

- Customer value metric + target
- Operator value metric + target
- Business value metric + target
- Compounding value (data/IP generated) + capture mechanism
- Moat dimension (which of the 4 is the primary defensible advantage)

### Section 3 — Process Map (Layer 1)

- Trigger(s)
- Nodes (name, owner, work-element list, latency budget, runtime mix)
- Edges (source, sink, carries, volume, latency, failure rate baseline)

- Queues (max depth, age limit, overflow handler)
- Boundaries (ownership, system, compliance)
- Sinks (happy-path + refuse + error + hard-refuse)
- Diagram (BPMN subset or mermaid)

## **Section 4 — Work Element Inventory**

For every work element: ID, type (1 of 12), input schema, output schema, latency budget, cost budget, failure mode + handler.

## **Section 5 — Runtime Assignments (*updated v1.1*)**

For every work element:

- Attribute scoring (decision class, action consequence, input type, volume × stakes)
- Constraint dimension scoring (data classification, provider contract, data residency, retention)
- Assigned runtime (D/C/A/H) + justification

## **Section 6 — Bounded Specs (*updated v1.1, for every C element*)**

- Enumerated action vocabulary
- Enumerated input classes
- OOD detector definition
- Explicit refusal triggers
- **Hard-refuse policy classes** (v1.1): list of input classes that bypass AI processing by policy

## **Section 7 — Grounding Specs (for every C element)**

- Data sources (with authority hierarchy)
- Freshness requirements
- Retrieval method
- Source ID requirement in output schema

## **Section 8 — Gate Specs (*updated v1.1, for every C element*)**

- Hard-refuse check rules
- Input gate rules
- Output gate rules (schema, policy, grounding, **confidence floor by action consequence**)
- Cross-check rules (**unconditional for irreversible-bounded**, otherwise conditional)
- Action gate rules (idempotency, rate, blast radius)

## **Section 9 — AI Step Specs (for every C element)**

- Model name + version (pinned)
- Prompt template (versioned, in git)
- Structured output schema
- Confidence definition + threshold ( $\geq$  floor for action consequence)
- Temperature, max tokens, other API parameters

## **Section 10 — Executor Specs (for every C element)**

- The deterministic code that takes AI output and performs the action
- Idempotency key
- Retry policy
- Rollback path

## **Section 11 — Telemetry + Observability**

- Events emitted per work element
- Required fields (input hash, output, model version, latency, cost, confidence, validator outcomes, error class)
- Storage location + retention
- Dashboard spec
- Control chart definitions
- Alarm routes (who/what gets paged, escalation timing)

## **Section 12 — Drift Spec**

- Input drift monitor (definition + threshold + response)
- Output drift monitor
- Confidence drift monitor
- Model version pinning policy
- Spec drift audit cadence

## **Section 13 — Governance Spec**

- Refusal taxonomy (graceful + hard-refuse, categorized)
- Fallback path
- Kill switch (who, conditions)
- Circuit breakers (per action class)
- Authority matrix (verb × threshold × who-or-what can execute)

## **Section 14 — Cost Model**

- Expected cost per run (broken down by work element)
- Routing logic (when expensive model triggered)
- Cost alarms (per-run, per-day, per-month)
- Quarterly re-bake plan

## **Section 15 — Change Management**

- Eval requirements before any change ships
- Canary plan (% of traffic, duration, success criteria)
- Rollback plan
- Approval requirements per change class

## **Section 16 — Ownership + Operations (*updated v1.1*)**

- Named owner (single human)
- On-call rotation
- Weekly trace-review ritual
- Monthly eval refresh



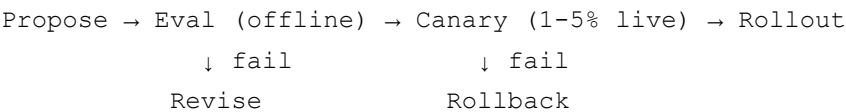
- Quarterly model + cost re-bake
- Incident response runbook
- **Terminal H status + reason class** (v1.1): for every H element, mark terminal/migration-eligible and reason

## Part V — Operations Discipline

### 5.1 Kaizen loop

| Cadence    | Activity                                                                 | Owner                    |
|------------|--------------------------------------------------------------------------|--------------------------|
| Continuous | Telemetry + alarms run automatically                                     | System                   |
| Daily      | Owner reviews andon-fired incidents from prior 24h                       | Owner                    |
| Weekly     | Owner reads 10-20 raw traces (Genchi Genbutsu); reviews drift dashboards | Owner                    |
| Monthly    | Eval set refresh; cost review                                            | Owner + AI engineer      |
| Quarterly  | Model re-bake                                                            | AI engineer              |
| Quarterly  | Spec audit (business rules, regulations, contracts changed?)             | Owner + business lead    |
| Quarterly  | Terminal H review (does the reason still hold?)                          | Owner + compliance       |
| Quarterly  | Architecture review (expand, change, retire workflow?)                   | You + supervisor council |

### 5.2 Change management path



| Change class | Eval requirement | Canary required | Approval |
|--------------|------------------|-----------------|----------|
|--------------|------------------|-----------------|----------|

|                                                    |                                      |              |                                  |
|----------------------------------------------------|--------------------------------------|--------------|----------------------------------|
| Prompt typo fix                                    | Gold set pass                        | No           | Owner                            |
| Prompt content change                              | Gold set $\geq$ baseline             | Yes, 24h     | Owner                            |
| Model version change                               | Full eval suite                      | Yes, 72h     | AI engineer + owner              |
| New work element                                   | Full eval + integration test         | Yes, 1 week  | Architecture review              |
| Runtime upgrade (A $\rightarrow$ C)                | Graduation criteria met              | Yes, 2 weeks | Architecture review              |
| Action vocabulary expansion                        | Full eval + safety review            | Yes, 2 weeks | Architecture review + you        |
| Hard-refuse class change                           | Full eval + compliance review        | Yes, 2 weeks | Architecture review + compliance |
| Constraint dimension change (BAA, residency, etc.) | Full eval + tool stack re-validation | Yes, 2 weeks | Architecture review + legal      |

### 5.3 Incident response

| Severity | Definition                                                      | Response time                           |
|----------|-----------------------------------------------------------------|-----------------------------------------|
| SEV-1    | Workflow producing wrong actions; customer or regulatory impact | <15 min: kill switch, route to fallback |
| SEV-2    | Validator-fail rate > 2x baseline for 1h+                       | <1 hour: investigate, throttle          |
| SEV-3    | Cost or latency anomaly without quality impact                  | <24h: investigate offline               |
| SEV-4    | Drift signal without immediate impact                           | Weekly review                           |

All incidents produce blameless postmortems: root cause, eval set additions, spec changes, action items.

### 5.4 Graduation criteria scale with per-error cost (v1.1)

Replaces v1.0's action-consequence-label-based thresholds. User calculates expected cost of a single wrong action (financial loss + customer harm + regulatory exposure + remediation cost), picks the appropriate band:

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Cost per error     | A → C agreement floor | Notes                                                         |
|--------------------|-----------------------|---------------------------------------------------------------|
| <\$10              | ≥95%                  | Standard reversible workflows                                 |
| \$10-\$100         | ≥99%                  | Most operational workflows                                    |
| \$100-\$10,000     | ≥99.5%                | Material business impact per error                            |
| \$10,000-\$1M      | ≥99.9%                | Often requires two-AI cross-check + permanent terminal H gate |
| >\$1M or unbounded | C not eligible        | Permanent terminal H                                          |

Per-error cost is declared in spec Section 16 and reviewed quarterly.

## Part VI — Worked Example #1: After-Hours Intake

### Context

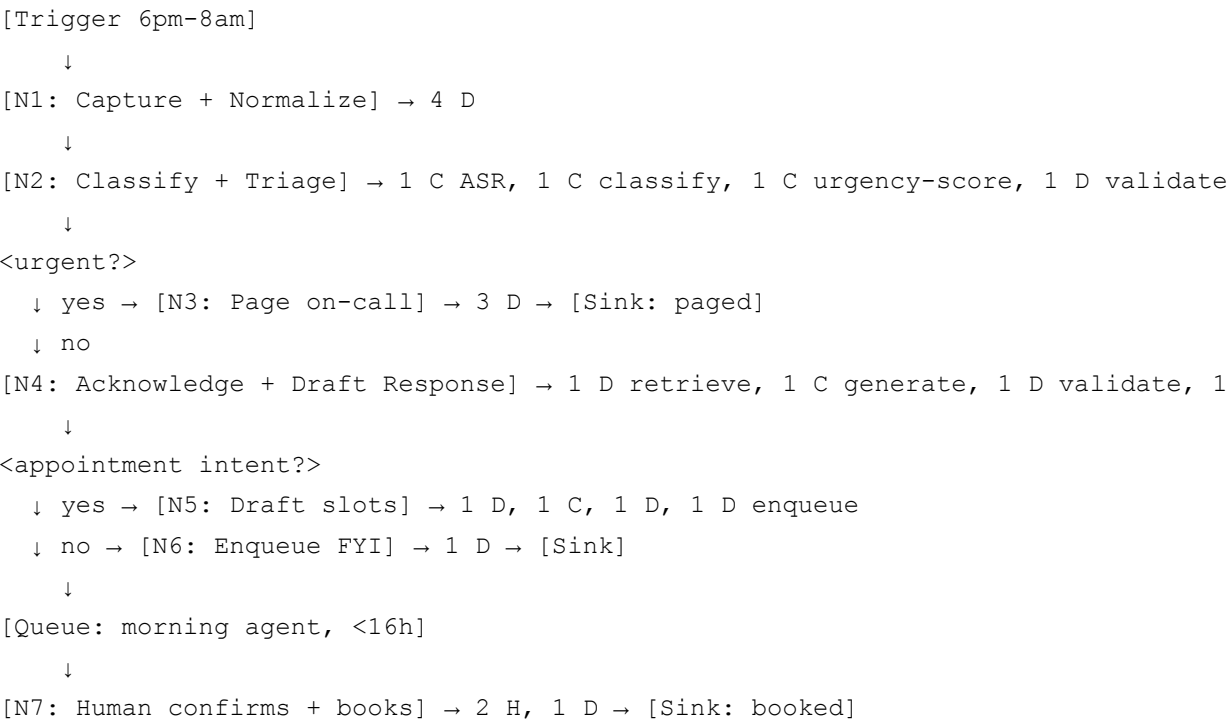
- Mode: **Replace** (eliminates the 87% who never get a callback)
- Volume: ~80/day across pilot clinics
- Stakes per run: low (reversible-costly at worst)
- Domain: Healthcare BPO (Classic Vision Care pilot)

### Section 2 — Value Promise

| Dimension   | Metric                                                 | Target                                      |
|-------------|--------------------------------------------------------|---------------------------------------------|
| Customer    | % contacts acknowledged within 90 sec                  | ≥95%                                        |
| Operator    | Morning agent picks up structured queue, not raw audio | 100%                                        |
| Business    | New-patient intent → booked appointment conversion     | ≥60% (baseline ~13%)                        |
| Compounding | Labeled intent + outcome rows per contact              | 1 per contact, feeds CARE/PAVP training set |

**Moat dimension:** Compounding. **Value mode:** Replace.

**Section 3 — Process Map (abridged)**



**Runtime distribution:** 18 D, 6 C, 3 H

**Section 5 — Constraint dimensions**

| Dimension           | Value                                                |
|---------------------|------------------------------------------------------|
| Data Classification | PHI (limited; intent + contact metadata)             |
| Provider Contract   | BAA required for any AI provider touching transcript |
| Data Residency      | US-only                                              |
| Retention           | 7-year HIPAA                                         |

**Section 6 — Bounded specs (wel-2.2 Classify Intent shown)**

- **Action vocab:** `new_patient_inquiry`, `existing_patient_appt_change`, `clinical_question`, `clinical_urgency`, `billing_question`, `general_inquiry`, `wrong_number`, `other`
- **OOD:** confidence <0.6 on top class OR language detection fails

- **Hard-refuse policy classes:**
  - Inputs from blocked numbers (deterministic check)
  - Test calls / known QA numbers (deterministic check)
  - Contacts from minors flagged in EHR (PHI lookup, route to morning queue with note)

## Section 8 — Gate specs (wel-4.2 Generate Ack shown)

- Hard-refuse: none additional beyond N2
- Input gate: triage object present, intent ∈ allowed enum
- Output gate: response ∈ approved template structures; confidence ≥0.8 (Reversible-costly default); no PHI beyond first name + clinic
- Cross-check: **conditional** (reversible-costly + low volume of risky cases = run cross-check when primary confidence <0.85)
- Action gate: send-text rate limit per number, dedup on contact\_id

## Section 16 — Terminal H

| H Element                         | Terminal? | Reason class                                                     |
|-----------------------------------|-----------|------------------------------------------------------------------|
| wel-7.1 Display proposal to agent | No        | Capability gap (will migrate to A in ~6 months)                  |
| wel-7.2 Agent confirms or amends  | No        | Capability gap (will migrate to A in ~6 months, C in ~18 months) |

## Per-error cost band

~\$50 per error (lost appointment booking + potential trust damage on bad text). Band: \$10-\$100. A→C threshold: ≥99% agreement.

## Cost model

| wel                 | Model        | \$/run   |
|---------------------|--------------|----------|
| 2.1 Transcribe      | provider ASR | \$0.006  |
| 2.2 Classify intent | Haiku        | \$0.0008 |
| 2.3 Score urgency   | Haiku        | \$0.0005 |

|                                             |        |                                  |
|---------------------------------------------|--------|----------------------------------|
| 4.2 Generate ack                            | Haiku  | \$0.0015                         |
| 4.4 Cross-check (conditional, ~30% of runs) | Sonnet | $\$0.005 \times 0.30 = \$0.0015$ |
| 5.2 Propose slots                           | Haiku  | \$0.002                          |
| <b>Total / contact</b>                      |        | <b>~\$0.012</b>                  |

## Part VII — Worked Example #2: Rx Fax → RevolutionEHR Entry

### Context

- Mode: **Augment** (cuts human transcription time; does NOT eliminate human approval)
- Volume: ~5/day per OD clinic
- Stakes per run: high (irreversible-bounded; clinical-liability exposure)
- Domain: Healthcare clinical operations (Classic Vision Care)

### Section 1 — Work Object

- **Definition:** One incoming Rx-related fax (new Rx, refill request, transfer) received at a Classic Vision Care fax endpoint from an outside prescriber, requiring entry into the patient's record in RevolutionEHR.
- **Examples:** new Rx fax from primary care, refill authorization fax from pharmacy, Rx transfer request from another optometrist.
- **In-scope:** All Rx-related faxes during business hours and after hours.
- **Out-of-scope:** Non-Rx faxes (handled by separate workflow), referral letters, lab results, controlled substances (hard-refuse, see §6).

### Section 2 — Value Promise

| Dimension | Metric                                                                                        | Target        |
|-----------|-----------------------------------------------------------------------------------------------|---------------|
| Customer  | Patient Rx record updated within 4 business hours of fax receipt                              | ≥95%          |
| Operator  | Staff time per fax reduced from ~12 min (manual entry) to ~3 min (AI extract + human approve) | 75% reduction |

|             |                                                                       |                                                                                   |
|-------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Business    | Eliminate transcription errors and reduce staff burden                | Zero clinical errors (target); 5-7 staff hours/week reclaimed across pilot clinic |
| Compounding | Labeled prescriber handwriting + Rx pattern dataset by prescriber NPI | Builds prescriber-quality lookup table over time                                  |

**Moat dimension:** Operator value (staff time reduction is the immediate ROI for the clinic owner). **Value mode:** Augment.

## Section 3 — Process Map

```

[Trigger: Fax received via BAA-covered fax service]
  ↓
[N1: Capture + Convert] → 4 wel (D): receive, store, dedupe, convert to image+tex
  ↓
[N2: Document Classification] → 3 wel: OCR (C), classify doc type (C), validate (
  ↓
<Rx-related?>
  ↓ yes
  ↓ no → [Route to non-Rx workflow]
[N3: Patient Identification] → 3 wel: extract IDs (C), lookup in EHR (D), score m
  ↓
<confident patient match?>
  ↓ yes
  ↓ no → [Refuse: queue for human ID]
[N4: Rx Field Extraction] → 7 wel: extract 6 fields (each C w/ high-gate), cross-
  ↓
[N5: Validation] → 4 wel: drug vs RxNorm (D), prescriber NPI lookup (D), controll
  ↓
<controlled substance?>
  ↓ yes → [HARD REFUSE: never auto-process, route to clinician immediately, audit
  ↓ no
[N6: Stage in EHR Pending Queue] → 1 wel (D)
  ↓
[Queue: pharmacist/RN/OD review queue]
  ↓
[N7: Human Review + Approve] → 2 wel (H, TERMINAL): review extraction (H), decide
  ↓
<approved?>
  ↓ yes
  ↓ no → [Sink: rejected, logged with reason, feeds eval set]
[N8: Write to EHR] → 1 wel (D): write Rx to RevolutionEHR via API (only after H a
  ↓

```

[N9: Archive] → 2 wel (D): archive fax + log completion

↓

[Sink: complete]

**Runtime distribution:** ~14 D, ~10 C, 2 H (both terminal)

## Section 5 — Runtime assignments + constraint dimensions

| Dimension           | Value                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| Data Classification | PHI + potential DEA-regulated (excluded by hard-refuse)                                                     |
| Provider Contract   | BAA required for every AI provider; <b>excludes Anthropic Computer Use, OpenAI Operator (not under BAA)</b> |
| Data Residency      | US-only                                                                                                     |
| Retention           | 7-year HIPAA + state pharmacy law                                                                           |

This pre-filters the tool stack: BAA-covered Bedrock (Claude on AWS) or Azure OpenAI for AI work. Excludes consumer ChatGPT, Anthropic Computer Use, OpenAI Operator.

## Section 6 — Bounded specs

**Hard-refuse policy classes (enforced before any AI):**

- Controlled substances (DEA Schedule II-V): detected by drug-name pattern match against DEA list; routed directly to clinician
- Pediatric Rx (patient age <18 at time of fax): hard-refuse, route to OD
- Inpatient/hospital prescriptions: hard-refuse, escalate
- Prescriber NPI not in approved list: hard-refuse, manual verification required first

**wel-4.2 (Extract drug strength + form) BOUNDED:**

- Output limited to {drug\_name, strength\_value, strength\_unit, form} from code-defined RxNorm-aligned vocabulary
- Free-text excluded
- OOD: confidence <0.92 (Irreversible-bounded floor) OR drug not found in RxNorm



## Section 7 — Grounding (wel-4.2)

- Sources: RxNorm API for drug normalization, NPI Registry for prescriber validation, clinic-specific formulary
- Required output field `source_region` = bounding box on original image showing where each field was read
- Required `rxnorm_id` validated against RxNorm API; output without it = reject

## Section 8 — Gate specs (wel-4.2)

- Hard-refuse: drug-name pattern check against DEA Schedule list before AI runs
- Input gate: image readable (resolution + clarity check), fax classified as Rx, patient matched  $\geq 0.95$
- Output gate: schema valid, RxNorm match required, confidence  $\geq 0.92$  (irreversible-bounded floor)
- Cross-check gate: **UNCONDITIONAL** (irreversible-bounded  $\rightarrow$  always cross-check); different OCR + LLM family on same image; mismatch on drug name or strength = refuse
- Action gate: no auto-write to EHR ever (Node 8 only fires after H approval at Node 7)

## Section 9 — AI Step Spec (wel-4.2)

- Model: claude-sonnet-4-6 (vision-capable, BAA-covered via Bedrock)
- Cross-check model: gpt-4-vision via Azure OpenAI (different family, BAA-covered)
- Temperature: 0
- Max tokens: 500
- Structured output schema:

```
{
  "drug_name": "<string from RxNorm>",
  "rxnorm_id": "<string>",
  "strength_value": "<float>",
  "strength_unit": "<enum: mg|mcg|g|mL|...>",
  "form": "<enum: tablet|capsule|liquid|...>",
  "confidence": "<float 0-1>",
  "source_region": "<bbox coords>"
}
```

## Section 11 — Telemetry

Per-fax events: {fax\_id, run\_id, wel\_id, started\_at, ended\_at, input\_hash, output, model\_version, cross\_check\_model\_version, cost\_usd, validator\_outcomes, refusal\_reason, prescriber\_npi, drug\_class}

Storage: S3 + Athena, 7-year retention.

Dashboards:

- Per-prescriber extraction accuracy (some handwriting is worse than others)
- Per-drug-class extraction accuracy
- Hard-refuse counts by class (DEA, pediatric, etc.) for compliance audit
- Cross-check disagreement rate
- Time-to-EHR-entry (lagging metric)

Control limits:

- Cross-check disagreement >5%: SEV-2
- Per-prescriber accuracy <90% for any prescriber: SEV-3, investigate, possibly add to manual-only list
- Hard-refuse false-negative rate (controlled substance missed) >0: SEV-1 immediate

## Section 13 — Governance

Refusal taxonomy:

- Hard-refuse: controlled\_substance, pediatric, inpatient, unrecognized\_prescriber
- Graceful refuse: low\_image\_quality, patient\_match\_uncertain, extraction\_low\_confidence, cross\_check\_disagreement

Authority matrix:

| Verb              | Auto? | Threshold              | Reviewer                        |
|-------------------|-------|------------------------|---------------------------------|
| ocr_extract       | Yes   | always                 | logged + cross-checked at wel-4 |
| classify_doc_type | Yes   | confidence $\geq 0.85$ | 100% sample (low volume)        |
|                   |       |                        |                                 |

|                        |                     |                                            |                   |
|------------------------|---------------------|--------------------------------------------|-------------------|
| extract_rx_field       | Yes                 | confidence $\geq 0.92$ + cross-check agree | 100% goes to N7   |
| validate_drug          | Yes                 | always                                     | deterministic     |
| stage_in_pending_queue | Yes                 | gate-pass only                             | 100% goes to N7   |
| <b>approve_rx</b>      | <b>No</b>           | <b>never</b>                               | <b>terminal H</b> |
| write_to_ehr           | Yes (deterministic) | only after H approval                      | logged            |

## Section 14 — Cost Model

| wel                                               | Model         | \$/run          |
|---------------------------------------------------|---------------|-----------------|
| 2.1 OCR                                           | Sonnet vision | \$0.012         |
| 2.2 Classify doc type                             | Haiku         | \$0.001         |
| 3.1 Extract patient IDs                           | Sonnet vision | \$0.008         |
| 4.x Extract Rx fields (6 fields, primary)         | Sonnet vision | \$0.012         |
| 4.x Cross-check (unconditional, different family) | GPT-4 vision  | \$0.015         |
| <b>Total AI cost / fax</b>                        |               | <b>~\$0.048</b> |

At ~5 faxes/day  $\times$  \$0.048 = ~\$0.25/clinic/day = ~~\$7.50/clinic/month. Negligible vs operator time savings (\$200/clinic/month in reclaimed staff hours).~~

## Section 16 — Ownership + Terminal H

- **Process owner:** Bre Dameron (operations leadership for all clinical workflows)
- **Clinical owner:** Pilot clinic OD
- **AI engineer:** TBD

### Terminal H declarations:

| H Element                 | Terminal?  | Reason class                                                                                                  |
|---------------------------|------------|---------------------------------------------------------------------------------------------------------------|
| wel-7.1 Review extraction | <b>Yes</b> | <b>Regulatory</b> (state pharmacy law + clinical liability) + <b>Risk Policy</b> (zero error tolerance on Rx) |
|                           |            |                                                                                                               |

|                               |            |                                                                                        |
|-------------------------------|------------|----------------------------------------------------------------------------------------|
| wel-7.2 Decide approve/reject | <b>Yes</b> | <b>Regulatory</b> (only licensed clinician can authorize Rx entry into patient record) |
|-------------------------------|------------|----------------------------------------------------------------------------------------|

Per-error cost band: \$10,000-\$1M (clinical error in Rx record = malpractice exposure + patient harm). A → C threshold: ≥99.9%. Combined with permanent terminal H = effectively "AI never approves; AI only proposes."

### Per-error cost reasoning

A wrong Rx in a patient's record could trigger: (a) wrong medication dispensed by pharmacy = patient harm, (b) malpractice liability for the practice, (c) board complaint to state optometry board, (d) regulatory action. Floor estimate \$10k-\$50k per occurrence; ceiling unbounded. Therefore: permanent terminal H at action time + ≥99.9% extraction accuracy required before AI is trusted to populate the proposal field.

---

## Part VIII — Evaluation Rubric (57 items)

This is the production-ready acceptance checklist. Walk it in order. Use for evaluating any existing flow or creating any new one.

### A. PROCESS-LEVEL CHECKS (Layer 1)

1. Work object defined as a single noun with explicit in/out scope?
2. Process drawn as a graph with trigger, nodes, edges, queues, decisions, sinks?
3. Every node has a single named human owner?
4. Every edge declares: latency budget, daily volume, baseline failure rate, data contract, trigger type?
5. Process has at least one happy-path sink, one refuse sink, and one hard-refuse sink?
6. Hidden queues (DLQs, retries) drawn on the map?
7. Implicit decisions made explicit as work elements?
8. No mega-nodes (10+ things)?

### B. WORK ELEMENT CHECKS

9. Each element is exactly one of 12 canonical types?
10. Each has one input class and one output class?

11. Required fields complete: type, schemas, latency budget, cost budget, failure mode?
12. Runtime (D/C/A/H) justified by attribute scoring?

### **C. RUNTIME ASSIGNMENT CHECKS (Layer 2)**

13. Scored: decision class, action consequence, input type, volume × stakes?
14. **(v1.1)** Scored: data classification, provider contract, data residency, retention?
15. Assigned the cheapest runtime satisfying all attribute and constraint dimensions?
16. C element passes the runtime eligibility matrix?
17. C element + irreversible-unbounded action → STOP, downgrade?
18. C element + ambiguous input → STOP, downgrade?
19. C element + novel or value-laden decision → STOP, downgrade?
20. **(v1.1)** All tools in the stack pass all four constraint dimensions?

### **D. CLOSED-LOOP PROPERTY CHECKS (for every C element)**

**BOUNDED** 21. Action vocabulary finite, enumerated, code-defined? 22. AI selects parameters but NEVER selects action class? 23. OOD detector with explicit threshold? 24. Explicit refuse path with logged reason? 25. **(v1.1)** Hard-refuse policy classes enumerated for this element's domain?

**GROUNDING** 26. Factual claims anchored to retrieved sources? 27. Output schema requires source IDs for factual fields? 28. Validator rejects ungrounded outputs?

**GATED** 29. **(v1.1)** Hard-refuse check runs before any other gate? 30. Input gate: schema + value ranges + content policy + OOD? 31. Output gate: schema + policy + grounding + **confidence ≥ floor for action consequence**? 32. **(v1.1)** Cross-check policy matches action consequence (unconditional for irreversible-bounded)? 33. Action gate: idempotency + rate + blast radius + sanity?

**OBSERVED** 34. Per-run telemetry: input hash, output, model version, latency, cost, confidence, validator outcomes? 35. Control charts on every key metric? 36. Input drift, output drift, confidence drift each monitored? 37. Champion-challenger on every model/prompt change?

**GOVERNED** 38. Refuse always available, cheap, logged? 39. Confidence thresholds explicit and version-controlled? 40. **(v1.1)** Hard-refuse classes audit-logged separately from graceful refuse? 41. Ambiguous inputs route to deterministic fallback? 42. Kill switch + circuit breakers per action class?

## **E. SPEC DOCUMENT COMPLETENESS**

43. All 16 sections present and complete?

## **F. COST DISCIPLINE**

44. Tiered routing (cheap first, escalate on low-confidence/high-stakes)?

45. Prompt caching for static system prompts?

46. Right-size retrieval with top-K cap?

47. Input gate refuses malformed before AI call?

48. Cross-check conditional where allowed by action consequence?

49. Rule-expressible work pushed to D not C?

50. Cost alarms per-run, per-day, per-month?

## **G. OPERATIONS**

51. Named owner with weekly trace-review ritual (10-20 raw traces)?

52. Monthly eval set refresh defined?

53. Quarterly model re-bake defined?

54. Incident response runbook with SEV-1/2/3/4 + response times?

55. Change management path (Propose → Eval → Canary → Rollout) with approvals per class?

56. **(v1.1)** Every H element declared terminal/migration-eligible with reason class?

57. **(v1.1)** Per-error cost band declared; graduation threshold matches band?

## **Verdict logic**

- All 57 pass → production-ready under AAC v1.1
- Any A/B/C fail → redesign process before build
- Any D fail on a C element → downgrade element to A or fix
- E fail → spec incomplete; not ready to hand to a builder
- F fail → workflow will hemorrhage cost; refactor before deploy
- G fail → workflow will silently decay; add operations discipline before deploy

## Use case framing

- **Evaluating an existing flow:** score 1-57, flag every failure, prioritize redesign
  - **Creating a new flow:** walk 1-57 in order; do not advance until current item passes
  - **Vendor/partner deliverables:** require this rubric as acceptance criteria
  - **Fractional TPL or AI engineer onboarding:** day-one reading
- 

## Appendix A — Glossary

| Term                 | Definition                                                                                     |
|----------------------|------------------------------------------------------------------------------------------------|
| Work element         | Atomic unit of work, one of 12 types                                                           |
| Node                 | Transactional grouping of work elements with one owner                                         |
| Process              | Full graph of nodes, edges, queues, decisions, triggers, sinks                                 |
| Runtime              | D / C / A / H — what executes a work element                                                   |
| Gate                 | Independent check that must pass before progression                                            |
| OOD                  | Out-of-distribution input                                                                      |
| Hard-refuse          | Policy-defined exclusion of an input class from AI processing, regardless of confidence        |
| Graceful refuse      | AI-driven exit when confidence/grounding inadequate                                            |
| Terminal H           | Human element that never migrates by default                                                   |
| Capability-gap H     | Human element that is migration-eligible ( $H \rightarrow A \rightarrow C$ )                   |
| Value mode           | Replace / Augment / Extend — how AI changes the human work                                     |
| Per-error cost band  | Expected cost of a single wrong action; determines graduation threshold                        |
| Constraint dimension | Data classification, provider contract, data residency, retention — pre-filters the tool stack |
| Refuse               | First-class outcome where AI declines and routes to deterministic fallback                     |
|                      |                                                                                                |

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| Champion-challenger | Running a new version in parallel with current production       |
| Canary              | Rolling out a change to a small % of traffic before full deploy |
| Andon               | Alert when a metric crosses a control limit                     |
| Genchi Genbutsu     | Lean discipline of reading raw traces, not aggregates           |
| Kaizen              | Continuous improvement loop                                     |
| Jidoka              | Stop-the-line on defect                                         |

## Appendix B — Cross-domain mapping of constraint dimensions

For users outside healthcare. Populate the 4 constraint dimensions per your domain:

| Domain             | Data Classification       | Provider Contract                | Data Residency           | Retention                        |
|--------------------|---------------------------|----------------------------------|--------------------------|----------------------------------|
| Healthcare (HIPAA) | PHI                       | BAA                              | US-only typically        | 7 years                          |
| Financial (US)     | NPI / PCI / MNPI          | DPA + sub-processor restrictions | Per regulation           | 7 years SOX, varies              |
| Financial (EU)     | Personal data under GDPR  | DPA + Article 28                 | EU-only                  | Per record type                  |
| Legal              | Privileged / work-product | Confidentiality agreement        | Per engagement           | Per matter + litigation hold     |
| Manufacturing      | Trade secret, ITAR        | NDA + export-control clauses     | Country-of-origin        | Indefinite for product liability |
| Education          | FERPA-covered             | Per institution policy           | Often state-specific     | Per student lifecycle            |
| Government         | FedRAMP / IL2-IL6         | FedRAMP-authorized only          | US-only, often gov cloud | Per record schedule              |
|                    |                           |                                  |                          |                                  |



|                      |     |     |                     |                       |
|----------------------|-----|-----|---------------------|-----------------------|
| Consumer /<br>Retail | PII | DPA | Per<br>jurisdiction | Per privacy<br>policy |
|----------------------|-----|-----|---------------------|-----------------------|

## Appendix C — What this replaces and what it doesn't

### Supersedes:

- AAC v1.0 (formerly "MSP-FLOW v1.0") and all v0.x drafts
- Ad-hoc AI workflow checklists
- Prompts + Lambda + dashboards without spec

### Does NOT replace:

- Make Sense Protocol v3.5 (separate sense-making framework; AAC and MSP are independent products)
- Dark Factory holdout QA system (this is the eval substrate that feeds AAC's evaluation layer)
- Two-agent pre-flight pattern (this is the cross-check gate in AAC Layer 3)
- Model Selection Harness (this is the quarterly re-bake mechanism)

## Appendix D — Pressure-test summary

AAC v1.1 has been pressure-tested against two structurally different workflows:

| Attribute           | After-Hours Intake  | Rx Fax → EHR               |
|---------------------|---------------------|----------------------------|
| Value mode          | Replace             | Augment                    |
| Volume/day          | ~80                 | ~5 per clinic              |
| Action consequence  | Reversible-costly   | Irreversible-bounded       |
| Input modality      | Audio + text        | Image (handwritten common) |
| Data classification | PHI (limited)       | PHI (full) + DEA-adjacent  |
| Hard-refuse         | 3 (minors, blocked, |                            |

|                     |                         |                                                          |
|---------------------|-------------------------|----------------------------------------------------------|
| classes             | test)                   | 4 (controlled, pediatric, inpatient, unknown prescriber) |
| Terminal H?         | No (migration-eligible) | Yes (regulatory + risk policy)                           |
| Per-error cost band | \$10-\$100              | \$10,000-\$1M                                            |
| Cross-check policy  | Conditional             | Unconditional                                            |
| AI cost/run         | ~\$0.012                | ~\$0.048                                                 |

The framework holds for both. The 8 v1.1 changes were surfaced by Rx fax and are now general-purpose.

Mental tests against three non-healthcare workflows (wire transfer review, contract redline, supplier RFQ) confirm the constraint dimensions and terminal-H reason classes populate cleanly across domains.

---

**End of AAC v1.1**